INTRODUCTION TO

Multivariate & Multivariable Analysis

STAT7242 ADVANCES BIOSTATISTICS

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LEARNING OBJECTIVES

01 Distinguish between multivariable and multivariate analysis

02 Understand why we control for confounding in public health research

03 Identify the correct regression model for different outcome types

04 Interpret key outputs: coefficients, odds ratios, confidence intervals

05 Recognize assumptions and limitations of regression methods

SESSION OUTLINE

- Foundations: Why Multivariable Analysis?
- Key Terminology & Conceptual Framework
- Types of Regression Models in Public Health
- Interpretation & Practical Application
- Assumptions, Limitations & Summary

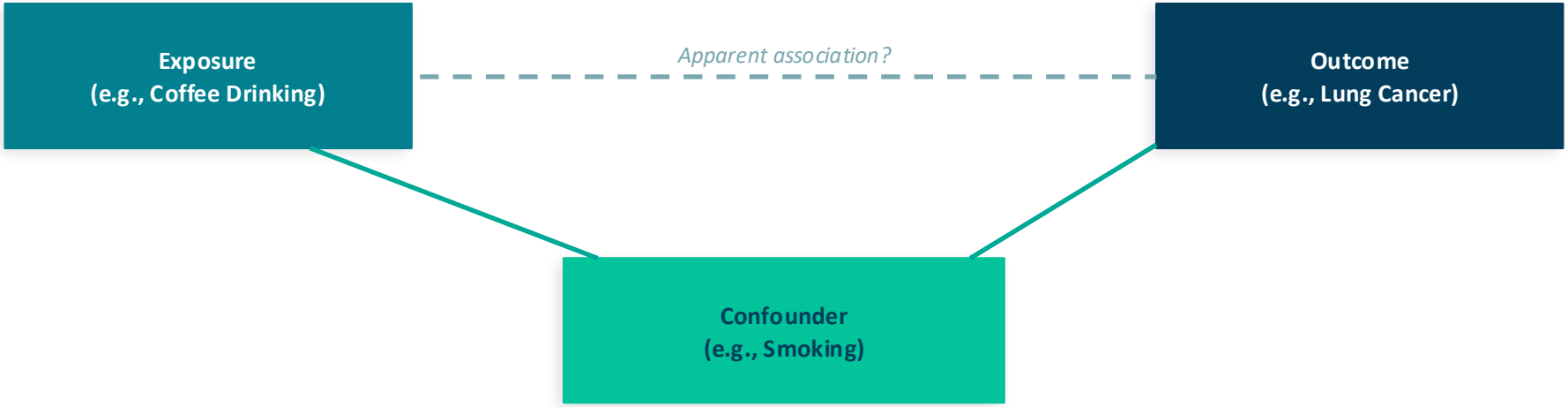
PART 1

Foundations

Why Multivariable Analysis?

The Core Problem: Confounding

Key insight: Crude (unadjusted) associations may be misleading without controlling for confounders.



Multivariable analysis allows us to isolate the effect of one variable while controlling for others.

Real-World Public Health Example

Crude (Unadjusted) Result

Study: Coffee intake & MI risk

OR = 2.1 (95% CI: 1.6–2.8)
"Coffee drinkers have 2× MI risk!"

⚠ But we have not adjusted for smoking, age, physical activity...

Adjusted (Multivariable) Result

After controlling for: *age, sex, smoking, BMI, physical activity*

OR = 1.05 (95% CI: 0.82–1.34)
"No significant association!"

✓ **Smoking was the true confounder**

Terminology: A Common Confusion

Multivariable Analysis

Multiple predictors → ONE outcome

- One dependent (outcome) variable
- Multiple independent (predictor) variables
- Most common in epidemiology & public health
- Example: Blood pressure ~ age + BMI + salt intake + smoking
- Methods: Multiple linear, logistic, Poisson regression

Multivariate Analysis

Multiple predictors → MULTIPLE outcomes

- Multiple dependent (outcome) variables simultaneously
- Less common in public health practice
- Example: [Diastolic, Systolic BP] ~ age + treatment
- Methods: MANOVA, factor analysis, cluster analysis, PCA
- Often used in psychological & social sciences

In this lecture, we focus on MULTIVARIABLE methods — the foundation of epidemiological analysis.

The Terminology Problem in Public Health

These two terms are frequently confused — even in peer-reviewed journals

"The terms multivariate and multivariable are not synonymous, yet they are frequently used interchangeably — this leads to analytical and interpretive errors."

— Adapted from Tabachnick & Fidell (2019), Ch. 1

MULTIVARIABLE

- One dependent variable (outcome)
- Multiple independent variables (predictors)
- Goal: explain or predict the single outcome
- Example: multiple linear regression, logistic regression
- Common in epidemiology & clinical research

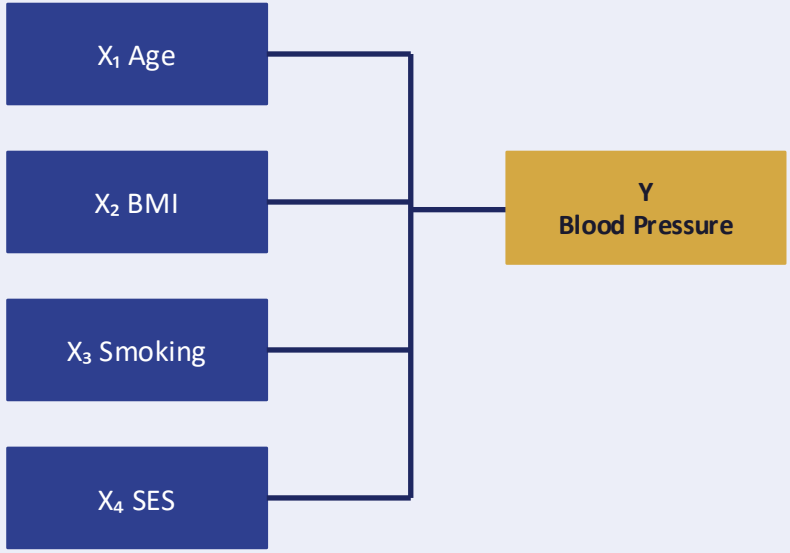
MULTIVARIATE

- Two or more dependent variables simultaneously
- Outcomes analysed together (not separately)
- Goal: model the joint distribution of DVs
- Example: MANOVA, SEM, factor analysis
- Common in psychology, social science, complex PH

A Visual Analogy: One River vs Many Streams

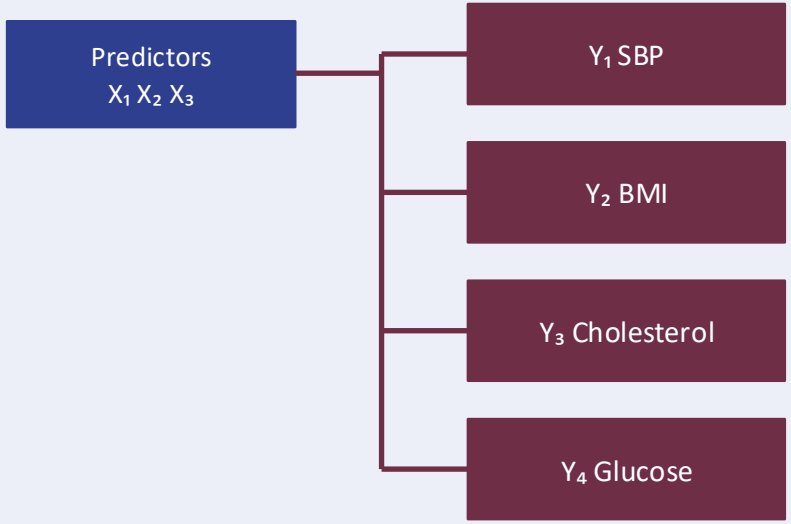
The number of outcome variables defines which class of analysis you need

MULTIVARIABLE



Many predictors → ONE outcome
Each IV assessed controlling for others

MULTIVARIATE



One set of predictors → MULTIPLE outcomes
Outcomes modelled jointly (not separately)

Side-by-Side Comparison

Feature	MULTIVARIABLE	MULTIVARIATE
Number of DVs	Exactly ONE	TWO or more
Number of IVs	Two or more (defines the 'multi')	One or more
Core question	What predicts Y?	What predicts Y_1, Y_2, … Y_k jointly?
Statistical model	Regression equation: $\hat{Y} = B_0 + B_1X_1 + \ldots$	System of equations or linear combination of DVs
Key techniques	Multiple regression, logistic regression, Cox	MANOVA, MANCOVA, SEM, factor analysis, cluster
Error control	Single α; adjust for multiple comparisons if needed	Controls experiment-wise α across multiple DVs
PH example	Predict SBP from age, BMI, smoking, SES	Compare interventions on SBP + BMI + cholesterol + glucose
Tabachnick ref.	Chapters 5–10 (regression family)	Chapters 7–14 (MANOVA, FA, DA, cluster, SEM)

Decision Guide: Multivariable or Multivariate?

How many outcome variables (DVs) does your research question have?

ONE outcome (DV)

→ MULTIVARIABLE ANALYSIS

- Continuous DV → Multiple Regression
- Binary DV → Logistic Regression
- Time-to-event → Cox Regression
- Count DV → Poisson / Neg. Binomial

TWO or MORE outcomes

→ MULTIVARIATE ANALYSIS

- Groups + continuous DVs → MANOVA/MANCOVA
- Latent structure → Factor Analysis
- Causal paths + latent → SEM
- Group classification → Discriminant/Cluster

Multivariable Analysis: Definition & Purpose

The most common analytical approach in clinical and epidemiological public health research

Multivariable analysis examines the relationship between ONE outcome variable and MULTIPLE predictor variables simultaneously. The goal is to estimate the independent effect of each predictor while statistically controlling for all others.

Confounding Control

Adjusts for variables that correlate with both exposure and outcome — essential for observational PH studies where RCT is not feasible

Effect Estimation

Quantifies the adjusted effect of each IV on the DV (B, β, OR, HR) after removing the influence of all other predictors in the model

Prediction

Builds a predictive model for clinical risk scores (e.g., Framingham Heart Score) combining multiple predictors optimally

Parsimony

Identifies which predictors add unique explanatory power — removes redundant variables to produce the simplest adequate model

Multivariable Techniques: The Regression Family

Choice depends on the measurement level of the single outcome variable

Technique	Outcome (DV)	Key Output	PH Application
Multiple Linear Regression	Continuous	B, β, R^2, F	Predict systolic BP from lifestyle factors
Binary Logistic Regression	Binary (yes/no)	Odds Ratio, 95% CI	Predict diabetes onset (yes/no)
Multinomial Logistic	Categorical ≥3 levels	OR for each category vs reference	Predict BMI category: normal/overweight/obese
Ordinal Regression	Ordered categories	OR (proportional odds)	Predict health rating: poor/fair/good/excellent
Poisson Regression	Count data	Incidence Rate Ratio	Predict ER visits per year
Cox Proportional Hazards	Time-to-event	Hazard Ratio, 95% CI	Predict time to CVD event controlling confounders
Negative Binomial	Overdispersed counts	IRR (adjusted for overdispersion)	Predict hospitalisation days

What Is Multivariate Statistics?

Definition (Tabachnick & Fidell)

Multivariate statistics refers to all statistical techniques that simultaneously analyse more than two variables.

These techniques extend univariate and bivariate methods to handle the complexity of real-world data involving many interrelated variables.

Why It Matters in Public Health

- Health outcomes are multifactorial — smoking, diet, SES, genetics interact simultaneously
- Control for confounders while testing multiple predictors
- Detect patterns across disease surveillance datasets
- Reduce dimensionality of complex survey instruments
- Classify patients into risk groups efficiently

Univariate → Bivariate → Multivariate

Univariate

Analysis of a single variable — describe or test its distribution

Examples:

- Mean BMI in a sample
- Prevalence of diabetes
- t-test: BMI in smokers vs non-smokers (one DV)

Bivariate

Relationship between exactly two variables

Examples:

- Correlation: BMI & blood pressure
- Chi-square: smoking × disease
- Simple linear regression

Multivariate

Simultaneous analysis of three or more variables

Examples:

- Multiple regression: BP ~ BMI + age + smoking + SES
- MANOVA: several outcomes at once
- Factor analysis of 30 survey items

Multivariate Analysis: Definition & Rationale

When outcomes are interrelated, analysing them jointly is more powerful and more valid

Multivariate analysis simultaneously analyses TWO OR MORE dependent variables. The joint analysis respects the correlational structure among outcomes and controls the experimentwise error rate (T&F, Ch. 7).

Why not just run multiple ANOVAs?

Running k separate tests inflates Type I error. With 4 outcomes at α=.05: P(at least one false positive) = $1-0.95^4$ = 18.5%. MANOVA keeps experiment-wise α at .05.

Respects outcome correlations

Outcomes are rarely independent (BP, BMI, cholesterol, glucose are all correlated). Multivariate methods model the covariance structure — this is more realistic.

Greater statistical power

When outcomes are correlated and effects are consistent across them, MANOVA has more power than multiple ANOVAs — the linear combination capitalises on shared variance.

Comprehensive group characterisation

A single multivariate test describes how groups differ across the whole outcome profile simultaneously, not piecemeal. Identifies the 'multivariate outlier group'.

Variables in Multivariate Analysis

Variable Type	Description	Public Health Example	Scale
Dependent (DV / Outcome)	Variable being explained or predicted	Systolic blood pressure, Disease status	Continuous or categorical
Independent (IV / Predictor)	Variable that explains the DV	Age, smoking status, physical activity	Any scale
Covariate	Continuous IV, often a confounder	Baseline BMI in an RCT	Continuous
Mediator	Explains the IV → DV pathway	Inflammation mediates stress → CVD	Any scale
Moderator	Modifies IV → DV relationship	Sex modifies drug × outcome	Any scale
Latent Variable	Not directly observed; measured via indicators	Socioeconomic status, stress	Unobserved construct

Multivariable

PART 2

Key Concepts & Conceptual Framework

Variable Types in Regression

Dependent Variable (Outcome)

e.g., Disease status, blood pressure, mortality, length of stay

Independent Variables (Predictors / Covariates)

e.g., Exposure of interest + all confounders you're adjusting for

Confounder

e.g., Associated with both exposure & outcome; not on causal pathway

Effect Modifier (Interaction)

e.g., Changes the magnitude of the exposure-outcome relationship

In public health regression: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k + \varepsilon$

SECTION 1

The "third variable" that distorts the true exposure–outcome relationship

What is a Confounder?

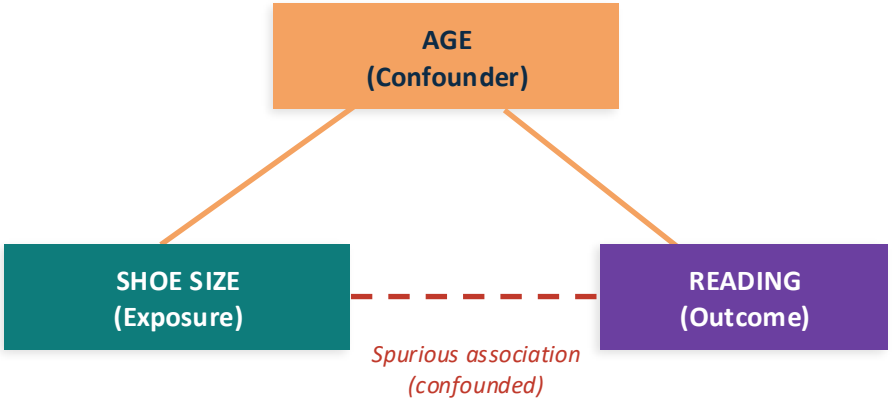
THREE CRITERIA FOR CONFOUNDING

① Associated with the EXPOSURE (but not caused by it)

② Associated with the OUTCOME (independently of exposure)

③ NOT in the causal pathway between exposure and outcome

CLASSIC EXAMPLE: Shoe Size & Reading Ability



CONTROL METHODS:

Design: Randomization, restriction, matching
Analysis: Multivariable regression, stratification, propensity scores

Detecting & Controlling Confounding

The 10% Rule

If adding a variable to a model changes the main exposure coefficient (β or OR) by ≥ 10%, it is considered a confounder and should be retained in the model.

Example: Coffee drinking → Lung cancer (crude vs. adjusted for smoking)

Model	OR (95% CI)	Interpretation
Crude (unadjusted)	2.4 (1.8 – 3.2)	Coffee appears strongly associated with lung cancer
Adjusted for smoking	1.1 (0.8 – 1.5)	Association disappears → Smoking was confounding!

Residual Confounding

Even after adjustment, confounding may persist due to imprecise measurement of the confounder, unmeasured confounders, or insufficient control. Acknowledge this as a study limitation.

SECTION 2

When the exposure–outcome relationship VARIES across subgroups

Moderation / Effect Modification

A moderator (M) is a variable that changes the STRENGTH or DIRECTION of the relationship between exposure (X) and outcome (Y). This is tested as a statistical interaction.

Testing for Interaction

Regression with product term:
$Y = \beta_0 + \beta_1 X + \beta_2 M + \beta_3 (X \times M)$

- β_3 = interaction coefficient
- Significant β_3 → effect modification present

Stratified analysis: Run separate models in each stratum of M and compare

Public Health Example

Does physical activity (X) reduce diabetes risk (Y)?

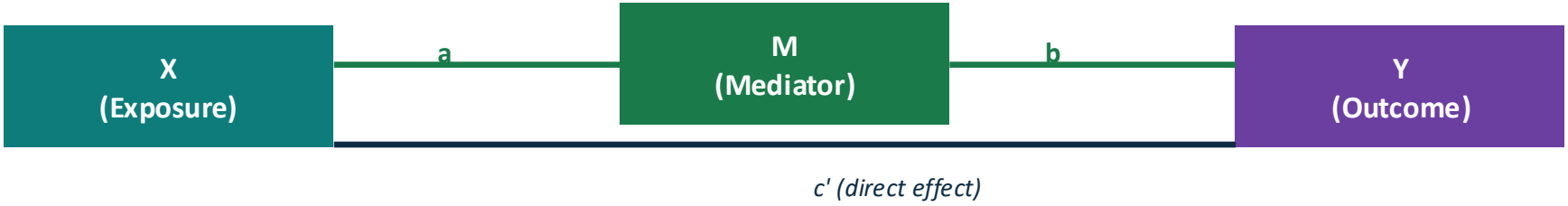
Moderator (M)	Effect of PA on DM Risk
Normal weight	OR = 0.6 (protective)
Obese	OR = 0.3 (strongly protective)

→ Weight status MODERATES the effect of physical activity. The interaction term (PA × BMI category) would be significant.

SECTION 3

Uncovering the MECHANISM through which exposure affects outcome

Mediation Analysis: Pathways & Methods



Indirect effect (mediated) = a × b

Total effect = direct (c') + indirect (a×b)

Baron & Kenny Steps

Step 1: X predicts Y (total effect)
Step 2: X predicts M (path a)
Step 3: M predicts Y controlling for X (path b)
Step 4: X effect on Y ↓ when M included (mediation)

Public Health Example

X: Low socioeconomic status
M: Poor diet quality (mediator)
Y: Type 2 diabetes

SES → Poor diet → Diabetes (indirect path)
SES → Diabetes directly (stress, access to care)

Confounder vs. Moderator vs. Mediator — At a Glance

CONFOUNDER	MODERATOR	MEDIATOR
Role	**Role**	**Role**
Distorts / biases	Modifies strength/direction	Explains mechanism
Position	**Position**	**Position**
Before both X and Y	Interacts with X	On causal pathway X→Y
How tested	**How tested**	**How tested**
Excluded/adjusted in model	Product term (X×M) in model	Path analysis / mediation
Analytic goal	**Analytic goal**	**Analytic goal**
Remove its effect	Describe for whom	Understand how/why
Example	**Example**	**Example**
Smoking confounds coffee→lung cancer	BMI moderates PA→diabetes	Diet mediates SES→diabetes

Confounding vs. Effect Modification vs. Mediation

Confounding

A third variable distorts the apparent exposure-outcome relationship

Action: CONTROL for it in regression

e.g., Smoking confounds coffee → lung cancer association

Effect Modification

The effect of exposure DIFFERS across strata of a third variable

Action: REPORT stratified results; include interaction term

e.g., Aspirin reduces MI risk more strongly in men vs women

Mediation

The third variable lies ON the causal pathway between exposure and outcome

Action: Do NOT adjust for mediators (blocks the causal effect)

e.g., Obesity mediates the relationship between diet and diabetes

PART 3

Types of Regression Models in Public Health

Choosing the Right Regression Model

Outcome Type	Distribution	Regression Model	Public Health Example
Continuous	Normal	Multiple Linear Regression	Predict systolic BP from age, BMI, salt
Binary (Yes/No)	Binomial	Logistic Regression	Odds of having diabetes given risk factors
Count data	Poisson	Poisson Regression	Number of asthma ER visits per year
Time-to-event	Survival	Cox Proportional Hazards	Time to death or disease remission
Ordinal categories	Ordinal	Ordinal Logistic Regression	Disease severity: mild/moderate/severe
Nominal categories	Multinomial	Multinomial Logistic Regression	Type of healthcare utilization

Types of Multiple Regression

Standard (Simultaneous)

All IVs entered at once. Each assessed as if entered last. Use when theory does not specify entry order. Most common.

Hierarchical (Sequential)

IVs entered in theoretically driven blocks. R^2 change at each step tests added predictive power. Preferred in confirmatory research.

Statistical (Stepwise/Forward/Backward)

Computer selects variables based on statistical criteria. Exploratory only — capitalises on chance. Requires cross-validation.

Logistic Regression

DV is categorical (binary/ordinal/multinomial). Estimates odds ratios. Standard in epidemiology for risk factor analysis.

Polynomial Regression

Adds squared or cubic terms to model curvilinear relationships. Centre variables first to reduce multicollinearity.

Interaction Terms

$X_1 \times X_2$ term tests moderation. If β is significant, the effect of X_1 depends on the level of X_2. Always include main effects.

Multiple Regression: Core Concepts

$$\hat{Y} = B_0 + B_1X_1 + B_2X_2 + \ldots + B_kX_k + \varepsilon$$

R^2

Proportion of variance in DV explained by all IVs combined.

Adjusted R^2 corrects for number of predictors.

B / β

B = unstandardised coefficient (units of X).

β = standardised (SD units) — compare relative importance across IVs.

sr^2

Squared semi-partial correlation — unique variance explained by each IV, controlling others. True unique contribution.

F & t

Overall model significance: F-test.

Individual predictor significance: t-test on each B coefficient.

Multiple Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k + \varepsilon$$

When to use:

Continuous outcome (e.g., SBP, BMI, length of stay, cost)

Goodness of fit:

R^2 — proportion of variance in outcome explained by model

Key assumption:

Outcome is normally distributed; linear relationship; homoscedasticity

Adjusted R^2:

Penalizes for adding unnecessary predictors

Interpretation:

β_1 = the change in Y for each unit increase in X_1, holding all other variables constant

Residuals:

ε = Y_observed − Y_predicted; should be normally distributed with mean 0

Example: SBP = 80 + 0.5(age) + 2.3(BMI) + 8.1(smoking) − 0.9(exercise_hrs/wk)

Multiple Linear Regression: Core Concepts

$$\hat{Y} = B_0 + B_1X_1 + B_2X_2 + B_3X_3 + \ldots + B_kX_k + \varepsilon$$

R^2	β	sr^2	VIF
Variance explained by all IVs combined	Standardised coefficient (SD units — compare IVs)	Unique variance of each IV (semi-partial r^2)	Variance Inflation Factor <5 acceptable; >10 serious

- **Hierarchical entry (T&F recommendation for confirmatory research):**
  - Enter IVs in theoretically motivated blocks; ΔR^2 at each step tests added predictive power
- **Standard (simultaneous) entry:**
  - All IVs entered at once; each assessed as if entered last — use when no a priori order
- *Avoid stepwise selection in confirmatory research — capitalises on sample-specific variance*

Logistic Regression — Most Common in Epidemiology

$$\log[P/(1-P)] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k$$

Outcome:	Binary (0/1): disease yes/no, death yes/no, outcome present/absent
Output:	Log-odds (logit), but we exponentiate βs to get Odds Ratios (OR)
OR interpretation:	OR=2.0 means the odds of the outcome are 2× higher per unit increase in X
Assumption:	Log-linear relationship between predictors and log-odds; no multicollinearity
Model fit:	Hosmer-Lemeshow test; AUC/ROC curve; pseudo-R^2 (Nagelkerke, Cox-Snell)

OR = e^β e.g., β = 0.693 → OR = e^0.693 = 2.0 (doubling of odds)

Logistic Regression: Binary Outcomes in Public Health

The standard multivariable technique when the outcome is categorical

The Model

Models the log-odds of the outcome:

$\log[p/(1-p)] = B_0 + B_1X_1 + B_2X_2 + \ldots$

Exponentiating B gives the Odds Ratio (OR) for each predictor controlling for others.

Key Outputs to Report

- OR with 95% CI for each predictor
- Wald χ^2 and p-value
- Nagelkerke R^2 (pseudo effect size)
- Hosmer-Lemeshow goodness-of-fit
- Classification accuracy (sensitivity/specificity)

Assumptions

- Linearity of log-odds (continuous IVs)
- No influential outliers
- No multicollinearity (VIF < 5)
- Independence of observations
- Adequate sample: min 10 events per IV

PH Example

Research Q: What predicts Type 2 Diabetes?
DV: Diabetes (yes=1 / no=0)
IVs: Age, BMI, family history, physical activity, diet score, SES
→ Adjusted OR for each after controlling all others

Poisson Regression & Cox Proportional Hazards

Poisson Regression

Use for: Count outcomes (number of events per person-time)

Formula: $\log(\text{count}) = \beta_0 + \beta_1 X_1 + \text{offset}$

Output: Incidence Rate Ratio (IRR)

Example: Hospital readmissions, ED visits, injury counts

Assumption: Mean = Variance (use negative binomial if overdispersed)

Cox Proportional Hazards

Use for: Time-to-event data (survival analysis)

Formula: $h(t) = h_0(t) \times e^{(\beta_1 X_1 + \beta_2 X_2)}$

Output: Hazard Ratio (HR)

Example: Time to death, remission, hospital discharge

Advantage: Handles censored observations (lost to follow-up)

Model Building Principles

1

Define Research Question

Clearly specify outcome (Y) and main exposure (X). State whether goal is prediction or causal inference.

2

Select Candidate Variables

Use subject-matter knowledge, directed acyclic graphs (DAGs), and prior literature. Avoid data-driven selection only.

3

Check Assumptions

Linearity, independence, homoscedasticity, normality of residuals (OLS). Check multicollinearity (VIF < 10).

4

Build & Assess Model

Crude model → adjusted model. Compare with likelihood ratio test, AIC/BIC, R^2, or pseudo-R^2.

5

Interpret & Report

Report coefficients, ORs, or HRs with 95% CI. Describe confounding adjustment and model fit.

PART 4

Interpretation & Practical Application

Reading a Regression Output Table

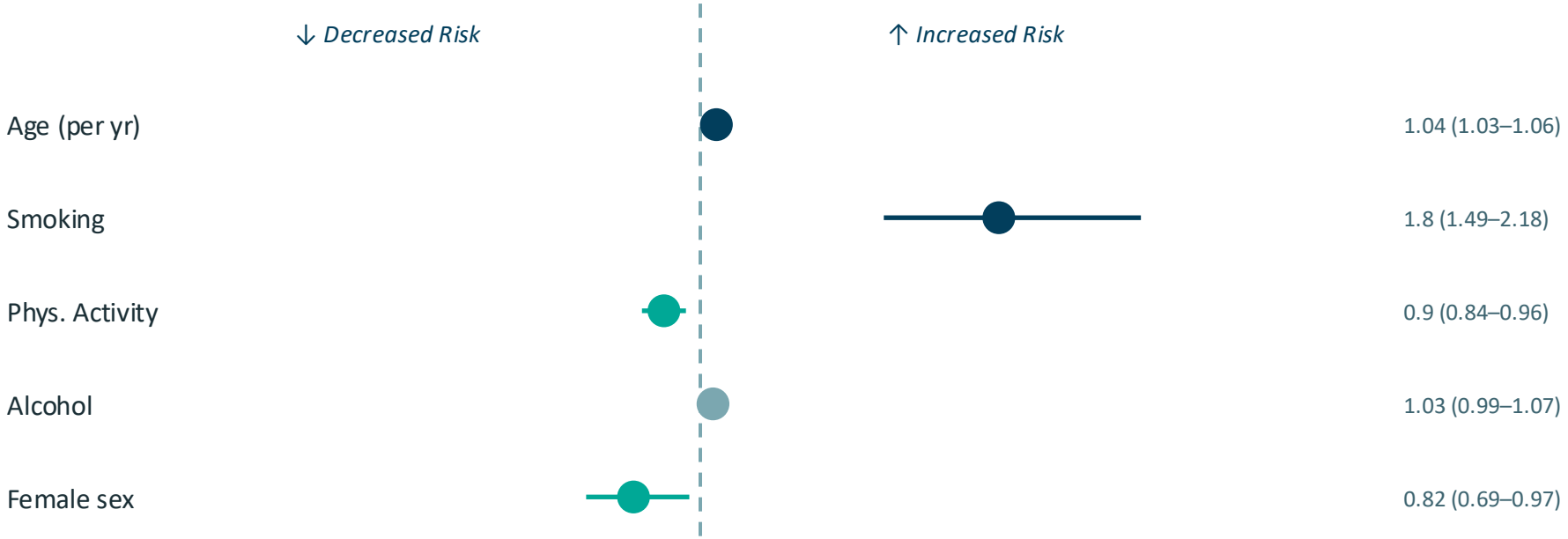
Outcome: Hypertension (Yes/No) — Logistic Regression, N=2,450 adults

Variable	β (Coef.)	SE	p-value	OR (95% CI)
Age (per year)	0.042	0.006	<0.001	1.04 (1.03–1.06)
BMI (per kg/m²)	0.071	0.012	<0.001	1.07 (1.05–1.10)
Smoking (yes vs no)	0.588	0.098	<0.001	1.80 (1.49–2.18)
Physical Activity (hrs/wk)	−0.110	0.035	0.002	0.90 (0.84–0.96)
Alcohol (drinks/wk)	0.025	0.019	0.19	1.03 (0.99–1.07)
Sex (Female vs Male)	−0.198	0.087	0.023	0.82 (0.69–0.97)

Alcohol p=0.19 — not statistically significant; teal bar = significant at p<0.05

Interpreting Odds Ratios & Confidence Intervals

Forest Plot (Visual Interpretation)



If 95% CI crosses 1.0 → not statistically significant at α=0.05 | Wider CI = less precise estimate

PART 5

Assumptions, Limitations & Summary

Key Assumptions of Regression Models

1 No Perfect Multicollinearity

Predictors should not be highly correlated with each other. Check VIF (>10 = problem). Multicollinearity inflates standard errors, making estimates unstable.

2 Correct Model Specification

The correct predictors are included. Omitting important confounders = omitted variable bias. Including irrelevant variables reduces precision.

3 Independence of Observations

Observations must be independent. Violations occur in clustered data (patients in hospitals), repeated measures, family-based studies. Use mixed models or GEE.

4 Adequate Sample Size

Rule of thumb: 10 events per variable (EPV) for logistic regression. Too few events = overfitting, unstable estimates, spurious associations.

Common Pitfalls in Multivariable Analysis

✗ Don't Do This	✓ Do This Instead
Adjusting for mediators	Identify causal pathway FIRST; do not adjust for variables on the causal pathway
Stepwise variable selection	Use theory-driven (a priori) variable selection based on DAGs and prior literature
Ignoring missing data	Report missingness patterns; use multiple imputation where appropriate
Extrapolation beyond data range	Interpret model predictions only within the range of observed data
Overfitting (too many variables)	Follow EPV rule; validate model on separate dataset or use cross-validation

Directed Acyclic Graphs (DAGs) — A Tool for Variable Selection

DAGs help us decide WHAT to adjust for — and what NOT to adjust for.

→ Adjust for confounders (common causes of both exposure & outcome)
→ Do NOT adjust for colliders (variables caused by both E and O)
→ Do NOT adjust for mediators (variables on the causal pathway)
→ Use DAGitty.net to draw and analyze your causal model

Key rules: (1) Adjust for confounders (2) Do NOT adjust for mediators (3) Do NOT adjust for colliders *(4) Use DAGitty.net to draw and verify model*

Reporting Multivariable Analysis — Best Practices

Reporting Checklist

- ✓ State the regression model used and justify the choice
- ✓ Report both unadjusted (crude) AND adjusted estimates
- ✓ List all variables included and rationale (DAG, literature)
- ✓ Report sample size, number of events, missing data
- ✓ Present effect estimates with 95% confidence intervals
- ✓ Report model fit statistics (R^2, AUC, Hosmer-Lemeshow)
- ✓ Describe how missing data were handled
- ✓ Conduct and report sensitivity analyses

Follow STROBE guidelines for observational studies | CONSORT for RCTs | TRIPOD for prediction models

Reporting Multivariable Analysis Results

What must appear in a well-written public health manuscript (APA 7 / STROBE guidelines)

Element	What to Report	Example Text
Sample & missing	N, % missing, imputation method	N=520; 3.1% missing, multiply imputed (m=20, Rubin's rules)
Assumption checks	Normality, VIF, outlier diagnostics	VIF range 1.2–2.4; 2 influential cases removed (Cook's D>1)
Model specification	Entry method, blocks, rationale	Hierarchical: Block 1 demographics, Block 2 lifestyle factors
Overall model fit	R^2 (linear); −2LL, AIC (logistic)	R^2=.42, Adj. R^2=.40, F(6,513)=61.4, p<.001
Individual predictors	B or OR, SE, 95% CI, β or Exp(B), p	BMI: B=1.24, SE=0.19, β=.33, 95% CI[0.87,1.61], p<.001
Effect size	sr^2, η^2, f^2, OR magnitude	sr^2=.09 for BMI; OR=2.14 for smoking (95% CI 1.52–3.01)
Power	A priori or post-hoc	Power=0.94 (G*Power; medium f^2=.15, α=.05, k=6)

Worked Example: Multivariable Logistic Regression

Research Question: What factors predict hypertension (BP ≥140/90) among adults 40–70 years? (N=652)

1	**Variable definition:**	DV: Hypertension (yes=1/no=0). IVs: Age (continuous), BMI (continuous), Smoking (yes/no), Salt intake (high/low), Physical activity (MET-min/wk), SES index (0–10).
2	**Data screening:**	Missing: 2.8% — MICE imputation. No multicollinearity (VIF max 2.1). 3 multivariate outliers excluded. Log-transform BMI (skewness=1.8).
3	**Model results:**	Age: OR=1.08/yr (1.05–1.11). BMI-log: OR=3.21 (2.14–4.82). Smoking: OR=1.74 (1.21–2.51). Physical activity: OR=0.97/MET (0.96–0.99). Salt: OR=2.08 (1.45–2.98).
4	**Model fit:**	Nagelkerke R^2=.39. Hosmer-Lemeshow $\chi^2(8)$=7.2, p=.51 (good fit). Overall classification: 79.3%. Sensitivity 74.1%, Specificity 82.6%.
5	**Conclusion:**	After adjusting for all covariates, BMI, age, smoking, salt intake, and low physical activity independently predicted hypertension. SES was non-significant after adjustment.

Summary: Choosing & Interpreting Your Model

Outcome Type	Model	Effect Measure	Example Public Health Use
Continuous	Multiple Linear	β coefficient	Mean difference in blood pressure
Binary	Logistic	Odds Ratio (OR)	Probability of disease given risk factors
Count/Rate	Poisson	Incidence Rate Ratio (IRR)	Rate of hospital admissions
Time-to-event	Cox PH	Hazard Ratio (HR)	Survival after cancer diagnosis
Ordinal	Ordinal Logistic	Cumulative OR	Severity score outcomes

All methods share the goal: isolate the effect of X on Y, holding all other variables constant (ceteris paribus).

Multivariate

Choosing the Right Multivariate Technique

A roadmap for matching your research question to the appropriate method

Tabachnick & Fidell's Selection Framework

Three key questions guide your choice of technique:

Q1: What is the nature of your research question?

Prediction? Group differences? Structure? Classification? Relationship?

Q2: How many DVs and IVs do you have?

One DV → regression family
Multiple DVs → MANOVA / discriminant
No clear DV → factor, cluster, MDS

Q3: What is the measurement level of your variables?

Continuous DV: regression, MANOVA
Categorical DV: logistic regression, discriminant
All categorical: log-linear

Major Multivariate Techniques at a Glance

Technique	# DVs	DV Scale	Primary Goal	PH Example
Multiple Regression	1	Continuous	Predict DV from IVs	Predict SBP from age, BMI, smoking
Logistic Regression	1	Categorical	Predict group membership	Predict disease yes/no
ANCOVA	1	Continuous	Group diffs controlling covariates	RCT: compare BP groups controlling baseline
MANOVA	≥2	Continuous	Multiple outcomes at once	Compare programs on BMI + cholesterol + BP
Discriminant Analysis	1 (group)	Categorical	Classify & find predictors	Predict high/medium/low risk
Factor Analysis	None	Continuous	Identify latent structure	Reduce 30 health survey items
Cluster Analysis	None	Any	Find natural groupings	Patient typologies
Survival Analysis	1 (time-to-event)	Time	Time to event	Time to rehospitalisation

Power & Sample Size in Multivariate Research

Adequate sample size is critical: multivariate analyses require more participants than univariate tests (T&F, Ch. 4).

80%	N ≥ 50+8k	N ≥ 200	20:1
Minimum acceptable power	Green's rule for regression (k = # IVs)	Minimum for factor analysis	Ideal cases-to-variable ratio

- **G*Power software:**
  - Free tool for calculating required N given α, power, effect size, and number of predictors
- **Cohen's f^2 effect sizes:**
  - Small = .02, Medium = .15, Large = .35 — medium effects need ~68 cases (5 IVs, α=.05, 80% power)
- **MANOVA:**
  - Requires more cases per cell; Pillai's trace is most robust to assumption violations and small n
- **Factor analysis:**
  - Communalities > .60 allow smaller N; weak communalities (< .40) may require N > 500

SECTION 02

Assumptions of Multivariate Analyses

Meeting the mathematical requirements that ensure valid results

Assumptions for Multivariate Analysis (T&F Ch. 4)

1. Multivariate Normality

Each DV and all linear combinations of DVs are normally distributed. Check: Mardia's skewness and kurtosis tests; Henze-Zirkler test. MANOVA robust when n per group ≥20.

2. Homogeneity of Variance-Covariance Matrices

Box's M test: covariance matrices should be equal across groups. Sensitive to non-normality — a significant M with large N may be tolerable. Use Pillai's trace if violated.

3. Linearity Among DVs

All pairs of DVs should be linearly related. Check scatterplot matrix (SPLOM). Violations reduce power of MANOVA — the linear combination cannot capture non-linear shared variance.

4. Absence of Multivariate Outliers

Mahalanobis D^2 for each case across all DVs; χ^2 critical value (df=# DVs, $p<.001$). Outliers strongly distort the multivariate centroid. Report n removed and sensitivity analysis.

5. Absence of Multicollinearity Among DVs

DVs should be correlated (that's the point) but not redundant ($r>.90$). Redundant DVs add noise without information. Check determinant of R matrix: should be >0.001.

6. Adequate Sample Size

N per cell > number of DVs; recommend n≥20 per cell. Overall N≥200 for SEM. Factor analysis: N≥300 or 10–20 subjects per variable. Power increases with group separation and fewer DVs.

Assumption 1: Normality

What It Means

Each variable (and the combination of variables — multivariate normality) should be approximately normally distributed in the population.

For large samples ($n > 200$), the Central Limit Theorem mitigates violations.

How to Check

- Histogram & Q-Q plots
- Skewness & kurtosis ($|z| > 3.29 = p<.001$)
- Shapiro-Wilk test (small n)
- Mardia's test (multivariate)

Remedies

- Log, square root, inverse transformations
- Winsorise extreme values
- Use robust or nonparametric alternatives
- Report sensitivity analyses

Assumptions 2 & 3: Linearity & Homoscedasticity

Linearity

Multivariate methods assume linear relationships among all pairs of variables.

Violation reduces power — the analysis cannot capture non-linear structure.

Homoscedasticity

Variance of residuals should be constant across levels of IVs (equal scatter).

In MANOVA: homogeneity of variance-covariance matrices (Box's M test).

Checking Linearity

Scatterplot matrix (SPLOM) of all variable pairs

Residuals vs. fitted values plot — look for non-linear trends

Checking Homoscedasticity

Residuals vs. fitted: look for funnel shape

Levene's test (univariate); Box's M (multivariate — sensitive to non-normality)

Assumption 4: Absence of Influential Outliers

Outliers distort parameter estimates and standard errors — they are among the most damaging assumption violations in multivariate analysis (T&F, Ch. 4).

3 Types	$z > \pm 3.3$	D^2/df	Cook's D
of outliers to consider	Univariate outlier flag ($p<.001$)	Mahalanobis distance cutoff	Influence on regression line

- **Univariate outliers**
  - Standardise each variable; flag $|z| > 3.29$
- **Multivariate outliers**
  - Mahalanobis distance: χ^2 with $p<.001$ flags cases influencing the centroid
- **Influential outliers**
  - Cook's D > 1 signals cases that substantially shift regression estimates
- Action: Describe, investigate, and consider running analyses with and without outliers (sensitivity analysis)

Assumption 5: Absence of Multicollinearity

The Problem

When predictors are highly correlated with each other ($r > .90$), the regression solution becomes unstable:

- Standard errors inflate dramatically
- Coefficients become unreliable and may flip sign
- The matrix inversion required by OLS becomes singular or near-singular

Singularity ($r = 1.0$) causes outright computational failure.

Detection

- Bivariate correlations > .90
- Variance Inflation Factor (VIF > 10 = problematic; > 5 = warning)
- Tolerance < .10 (= 1/VIF)
- Condition Index > 30 signals multicollinearity

Solutions

- Remove or combine redundant predictors
- Compute a composite score (e.g., sum or factor score)
- Ridge regression (penalises large coefficients)
- Principal Component Regression

Data Screening Before Analysis (T&F, Ch. 4)

Tabachnick & Fidell recommend a systematic screening protocol before any multivariate analysis:

01

Accuracy

Check data entry errors, range checks, frequency distributions for impossible values

02

Missing Data

Assess pattern (MCAR/MAR/MNAR), compute % missing, choose strategy (MI, FIML, deletion)

03

Outliers

Univariate (z-scores), multivariate (Mahalanobis D^2), influential (Cook's D)

04

Normality

Histograms, skewness/kurtosis, Q-Q plots; transform if needed

05

Linearity

Scatterplot matrix; curvilinear relationships reduce power

06

Homoscedasticity

Residual plots; Box's M for MANOVA; Levene's for ANOVA

Missing Data: Patterns & Strategies

MCAR — Missing Completely At Random

Missingness is unrelated to any variable (observed or unobserved). Listwise deletion valid but wastes data. Rare in practice.

MAR — Missing At Random

Missingness is related to observed (not missing) variables. Multiple Imputation (MI) and Full Information Maximum Likelihood (FIML) are unbiased.

MNAR — Missing Not At Random

Missingness depends on the unobserved value itself (e.g., depressed patients skip depression items). Requires selection models or sensitivity analysis.

Recommended Strategy

Use MI with 5–20 imputed datasets (Rubin's rules). FIML in SEM. Avoid pairwise deletion. Document all decisions and threshold (e.g., >5% triggers investigation).

SECTION 04

Analysis of Variance & MANOVA

Testing group differences on one or multiple outcomes simultaneously

ANCOVA and MANOVA in Public Health

ANCOVA

- Extends ANOVA by controlling one or more continuous covariates
- Removes covariate variance before testing group differences → increases power
- Essential in RCTs: compare post-test means controlling for baseline
- Assumption: Homogeneity of regression slopes (covariate × group interaction non-significant)
- PH Example: Compare 3 exercise programs on HbA1c controlling for baseline HbA1c and age

MANOVA

- Tests group differences on multiple DVs simultaneously
- Controls Type I error inflation from running multiple ANOVAs
- Creates a linear combination of DVs (variate) that maximises group separation
- Multivariate test statistics: Wilks' Λ, Pillai's V, Hotelling's T^2, Roy's largest root
- Follow up significant MANOVA with univariate ANOVAs and discriminant analysis
- PH Example: Compare urban vs rural vs semi-urban on BMI + BP + cholesterol + glucose

MANOVA: Multivariate Analysis of Variance

The multivariate extension of ANOVA — tests group differences on multiple DVs simultaneously

How MANOVA Works

- Forms a linear combination (variate) of DVs that maximises group separation
- Tests whether group centroids (multivariate means) differ significantly
- Produces FOUR test statistics — each has advantages in different situations
- Follow-up: univariate ANOVAs and discriminant analysis to interpret which DVs drive differences
- Requires: homogeneity of variance-covariance matrices (Box's M test) and multivariate normality

The Four Test Statistics

Pillai's Trace (V)

Most robust; recommended when N is small or assumptions are violated. Use by default.

Wilks' Lambda (Λ)

Most widely reported. Smaller Λ = larger group difference. Multivariate equivalent of F-ratio.

Hotelling's Trace (T^2)

Powerful when groups are separated on one dimension. Use with 2 groups.

Roy's Largest Root

Most powerful if one variate explains most variance; liberal when assumptions violated.

MANCOVA & Profile Analysis

Extensions of MANOVA for covariate adjustment and repeated measures

MANCOVA

Extends MANOVA by statistically controlling one or more continuous covariates.

Use when groups differ on a covariate that affects all DVs (e.g., baseline values in an RCT).

Assumption: Homogeneity of regression — covariate must have the same relationship to DVs across groups (test the group × covariate interaction).

Profile Analysis

Special case of MANOVA for repeated measures or multiple similar DVs (same construct, different times/conditions).

Tests three hypotheses:
1. Parallelism: Do group profiles run parallel?
2. Equal levels: Are overall profiles at same height?
3. Flatness: Is the profile flat (no change)?

PH Example: Compare male vs female profiles of mood, anxiety, fatigue, and pain scores over 4 measurement occasions.

Assumptions: MANCOVA

- Multivariate normality (Mardia's test)
- Homogeneity of variance-covariance matrices (Box's M)
- Linear relationship between covariates and DVs
- Independence of covariate and group membership
- Sufficient n per cell (recommend ≥20)

Effect Sizes to Report

- η^2p (partial eta-squared) for each MANOVA effect
- Cohen's d for follow-up univariate contrasts
- Wilks' Λ converted to F (partial $\eta^2 = 1 - \Lambda^{(1/s)}$)
- Multivariate effect size: η^2mult
- Report 95% CI for all effect sizes

Worked Example: MANOVA in Public Health

Research Question: Do three community health programs differ on a cardiovascular risk profile? (N=285, 95 per group)

1	**Variables:**	IV: Program group (A=Exercise, B=Diet, C=Combined). DVs: SBP, BMI, Total Cholesterol, Fasting Glucose (4 continuous outcomes). Covariate: Baseline BMI.
2	**Data screening:**	Mardia's test: $p=.12$ (normality OK). Box's M: $F(30,198)=1.41$, $p=.08$ (homogeneity met). 2 multivariate outliers (D^2) removed. Determinant of $R=0.14$ (no redundancy).
3	**MANCOVA result:**	Pillai's $V=0.38$, $F(8,556)=14.7$, $p<.001$, $\eta^2p=.17$. Groups differed significantly on the combined DV profile after controlling baseline BMI.
4	**Follow-up ANOVAs:**	SBP: $F(2,282)=18.3$, $p<.001$, $\eta p^2=.11$. Cholesterol: $F(2,282)=22.1$, $p<.001$. BMI: $F(2,282)=9.4$, $p<.001$. Glucose: $F(2,282)=3.1$, $p=.047$.
5	**Post-hoc & conclusion:**	Tukey HSD: Combined > Exercise > Diet on all outcomes. Combined program produced the largest improvement across the full cardiovascular profile. Report all 4 DVs, not just significant ones.

SECTION 05

Factor Analysis & Data Reduction

Uncovering hidden structure in public health surveys and instruments

Factor Analysis: EFA vs CFA

Exploratory Factor Analysis (EFA)

Discover underlying factor structure — no prior theory needed.

Extraction: Principal Axis Factoring (T&F recommendation) or Maximum Likelihood.

Rotation: Orthogonal (Varimax) if factors unrelated; Oblique (Promax, Oblimin) if correlated.

Confirmatory Factor Analysis (CFA)

Test a pre-specified model against data.

Requires structural equation modelling software (AMOS, lavaan in R, Mplus).

Fit indices: CFI > .95, RMSEA < .06, SRMR < .08 indicate adequate fit.

Key Decisions in EFA

How many factors? → Scree plot, parallel analysis, eigenvalue > 1 (Kaiser rule, but weak)

What loading threshold? → |.32| minimum; |.45| preferred for interpretation

Delete cross-loading items (load >.32 on 2+ factors)

Public Health Application

Scale development: reduce 40 perceived-stress items to 3 underlying dimensions.

Item reduction: shorten health literacy instrument while preserving construct validity.

Factor scores can become predictors in subsequent regression.

Cluster Analysis & Discriminant Analysis

Cluster Analysis

- Groups observations (not variables) into homogeneous clusters based on profile similarity
- Hierarchical: Ward's method (minimises within-cluster variance) — use dendrogram to choose k
- Non-hierarchical: k-means — requires specifying k in advance; use hierarchical solution to seed
- Validate clusters: ANOVA on cluster means; discriminant analysis to assess separation
- PH Example: Identify patient typologies (e.g., multimorbidity clusters) for targeted interventions
- Cluster membership used as grouping variable in subsequent analyses

Discriminant Analysis

- Identifies variables that best discriminate between known groups; classifies new cases
- Forms discriminant functions (variates) — each a weighted combination of IVs
- Can follow MANOVA: which DVs account for group separation?
- Jackknifed (leave-one-out) classification: unbiased accuracy estimate
- PH Example: Predict HIV risk category (low/medium/high) from behavioural and social predictors
- Logistic regression preferred when assumptions are severely violated

Factor Analysis & Structural Equation Modelling

Latent variable multivariate techniques for instrument development and causal modelling

Factor Analysis (EFA/CFA)

- **EFA (Exploratory):**
  - Discover latent structure in a set of variables. No prior theory required. Use Principal Axis Factoring + Promax rotation (T&F recommendation)
- **CFA (Confirmatory):**
  - Test a theoretically specified model. Requires SEM software (lavaan, Mplus, AMOS). Report CFI>.95, RMSEA<.06, SRMR<.08
- **Loading threshold:**
  - |.32| minimum; |.45| preferred. Delete cross-loading items. Communality <.30 = poor item
  - PH use: Reduce 30 SF-36 items to latent health dimensions for use as DVs in MANOVA

Structural Equation Modelling (SEM)

- **Simultaneously estimates:**
  - Measurement model (CFA: latent → observed) AND structural model (latent → latent paths)
  - Can test mediation, moderation, and complex causal pathways in a single model
- **Model fit:**
  - Report CFI, TLI, RMSEA (with 90% CI), SRMR, χ^2 (with df and p)
  - N requirements: minimum 200; prefer 10–20 observations per free parameter
  - PH use: Test whether SES → perceived stress → inflammatory markers → CVD risk; mediation with latent variables

Discriminant & Cluster Analysis

Classification and grouping techniques in the multivariate family

Discriminant Analysis (DA)

Identifies which variables best discriminate between known groups; classifies new cases.

Forms discriminant functions (weighted composites of IVs) that maximise group separation.

Follow-up to MANOVA: which DVs account for the significant multivariate effect?

PH Example: Classify patients as low/medium/high CVD risk from

Cluster Analysis

Groups cases (not variables) into clusters based on similarity across multiple variables.

Hierarchical (Ward's method) → use dendrogram to choose k. Then k-means to refine.

Validate: ANOVA on cluster means; DA to assess separation quality.

PH Example: Identify patient typologies (e.g., lifestyle clusters)

Key DA Assumptions

- Multivariate normality within groups
- Homogeneity of covariance matrices
- No extreme outliers (Mahalanobis D^2)
- Linear relationships among predictors
- Cross-validate: jackknifed classification accuracy

Key Cluster Considerations

- Standardise all variables before clustering (z-scores)
- Choice of distance measure (Euclidean vs squared Euclidean)
- Solution is not unique — must be validated externally
- Report silhouette coefficients or Calinski-Harabasz index
- Cluster membership = IV in subsequent regression

SECTION 06

Interpretation, Reporting & Critical Appraisal

Reading and writing multivariate results with confidence

Reporting Multivariate Results (APA 7 Standards)

Element	What to Report	Example
Sample	N, missing data handling, inclusion/exclusion criteria	N = 342; 4.2% missing; multiply imputed (m = 20)
Assumption checks	Normality, outliers, collinearity, homoscedasticity	VIF range 1.3–2.1; 3 multivariate outliers removed
Model fit	R^2, Adjusted R^2, F, df, p for regression; Wilks' Λ for MANOVA	$R^2 = .47$, Adj. $R^2 = .44$, $F(5, 336) = 59.7$, $p < .001$
Coefficients	B, SE, β, t/z, p, 95% CI for each predictor	BMI: B = 1.23, SE = 0.18, β = .34, $p < .001$, CI[0.88, 1.58]
Effect sizes	R^2, sr^2, η^2, ω^2, Cohen's d, OR	$sr^2 = .08$ for BMI; $\eta p^2 = .12$ for group effect
Power	Achieved power or priori power analysis	Post-hoc power = 0.92 (G*Power 3.1, medium f^2)

Critically Appraising Multivariate Studies

When reading published multivariate analyses, ask these questions:

Technique match

Is the technique appropriate for the research question and variable types?

Sample size

Is N adequate for the number of predictors/factors? Was power estimated?

Assumption reporting

Were normality, linearity, outliers, and collinearity examined and reported?

Missing data

Was missingness described? Was an appropriate strategy (MI, FIML) used?

Overfitting

Was the model validated (cross-validation, holdout sample, shrinkage)?

Effect sizes

Are effect sizes reported alongside p-values? Are they practically meaningful?

Causality claims

Does the design support causal inference? Are cross-sectional limitations acknowledged?

Replication

Are enough details provided to replicate? Is data/code available?

Common Pitfalls in Multivariate Research

Fishing for significance

Running many analyses and reporting only significant results inflates Type I error. Pre-register hypotheses.

Ignoring assumptions

Proceeding without screening data leads to biased estimates. Document and address each violation.

Conflating statistical & practical significance

With large N, trivial effects become significant. Always report R^2, η^2, OR alongside p-values.

Overfitting with stepwise methods

Computer-selected models capitalise on chance variation. Replicate in independent samples.

Interpreting regression as causal

Cross-sectional regression cannot establish causality. Use causal language carefully.

Underreporting methods

Readers cannot critique what they cannot see. Report all decisions: screening, transformations, outlier handling.

Worked Example: Predicting Hypertension Risk

Research Question: Which modifiable factors predict systolic blood pressure in adults aged 40–70?

1	**Study Design & Variables:**	Cross-sectional survey, N = 520. DV: Systolic BP (continuous). IVs: BMI, age, smoking (y/n), physical activity (MET-min/wk), sodium intake, SES score.
2	**Data Screening:**	Missing: 3.2% — multiply imputed. 2 multivariate outliers removed. BMI & sodium log-transformed (skewness). VIF max 2.4 — no multicollinearity.
3	**Analysis: Hierarchical Regression:**	Block 1: Demographics (age, sex) — $R^2 = .21$, $F(2,517) = 68.7$, $p<.001$. Block 2: + lifestyle IVs — $\Delta R^2 = .19$, $\Delta F(4,513) = 31.2$, $p<.001$. Final $R^2 = .40$.
4	**Results & Interpretation:**	Age (β=.31), BMI-log (β=.28), sodium (β=.22) and smoking (β=.14) all significant. Physical activity β=-.18, p<.001 (protective). SES ns after adjustment.
5	**Reporting:**	All β reported with 95% CI, sr^2, p-values. Achieved power = .97. Sensitivity analysis excluding outliers yielded identical conclusions.

Choose wisely

Match technique to research question, DV type, and number of variables

Screen first

Always screen data for outliers, normality, collinearity, and missing data before analysis

Report fully

Assumptions, effect sizes, CIs, power — not just p-values

Think causally

Statistical control ≠ causal control; acknowledge design limitations

Validate models

Cross-validate, replicate; be sceptical of stepwise-selected models

Keep reading T&F

Tabachnick & Fidell remains the definitive practical reference — consult specific chapters as needed

Reference: Tabachnick, B.G. & Fidell, L.S. (2019). Using Multivariate Statistics (7th ed.). Pearson.

Key Take-Home Messages

01 Multivariable ≠ Multivariate. In public health, we almost always mean multivariable (one outcome, many predictors).

02 The purpose of multivariable analysis is to obtain unconfounded, adjusted estimates of association.

03 Choose your model based on the outcome type: linear for continuous, logistic for binary, Poisson for counts, Cox for survival.

04 Always draw a DAG before running your model to guide variable selection and avoid bias.

05 Report both crude and adjusted estimates; interpret ORs, IRRs, and HRs in plain language with confidence intervals.

Choose wisely

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Assumptions, effect sizes, CIs, power — not just p-values

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Reporting Standards: What Must Be Included

Both STROBE (observational PH) and APA 7 guidelines — applying to multivariable and multivariate equally

Reporting Element	Multivariable (e.g., Logistic Regression)	Multivariate (e.g., MANOVA)
Sample	N, % missing, imputation method	N per cell, % missing, imputation
Assumptions	Normality of residuals, VIF, Cook's D	Mardia's, Box's M, D^2, linearity of DVs
Model specification	IVs, blocks, entry method	DVs, IVs, covariates, design type
Overall test	−2LL, AIC, χ^2, R^2 (Nagelkerke)	Pillai's V (preferred), F, df, p
Effect size	OR + 95% CI, Nagelkerke R^2	η^2p per effect; η^2 per follow-up ANOVA
Individual effects	B, SE, Wald χ^2, p, 95% CI	F, df, p, η^2p per DV in follow-up
Power	A priori or post-hoc (G*Power)	A priori or post-hoc (G*Power)
Sensitivity analysis	Results with/without outliers	Results with Wilks' Λ; outlier sensitivity

Questions & Discussion

Recommended Resources

- Vittinghoff E et al. Regression Methods in Biostatistics (2nd ed.)
- Hosmer DW & Lemeshow S. Applied Logistic Regression (3rd ed.)
- Greenland S et al. Statistical tests, P values, confidence intervals. Eur J Epidemiol 2016
- DAGitty: dagitty.net — Draw and analyze causal diagrams online
- Kelley DL. STROBE Checklist — strengthening observational research reporting